Digital Eye Strain Detection System

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Abstract

In the contemporary era of prolonged digital exposure, Computer Vision Syndrome (CVS) is characterized by eye strain, dryness, and fatigue due to which it has emerged as a significant public health concern. This project presents the design and implementation of a real-time monitoring system that utilizes computer vision to tackle these effects. By tracking the Eye Aspect Ratio (EAR) for blink frequency and facial geometry for screen proximity, the system provides an automated, non-intrusive solution to promote healthier digital habits.

Introduction

Digital Eye Strain is caused by two primary factors:

1. Reduced Blink Rate: Humans typically blink 15–20 times per minute, but this drops by nearly 60% when focusing on a screen, leading to corneal dryness.

2. Improper Viewing Distance: Sitting too close to a monitor increases the accommodative demand on the eyes and contributes to neck and shoulder tension.

Problem Statement

Most users are unaware of their decreasing blink rates during intense work sessions. There is a need for a lightweight, software-based tool that can run in the background and alert users when their habits become hazardous to their ocular health. Chronic digital eye strain can lead to long-term ocular health issues, including permanent damage to the tear film and increased risk of early-onset myopia.

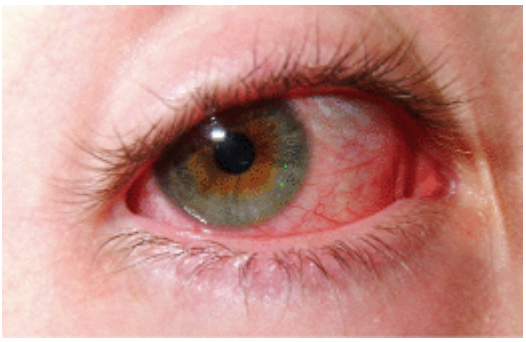


Fig 1: Representation of eye strain

Objectives

• Develop a real-time blink detection algorithm using facial landmarks.

• Implement a proximity alert system to maintain an ergonomic distance from the screen.

• Ensure the system is computationally efficient enough to run on standard consumer hardware.

System Architecture and Design

- Facial Landmark Localization

The system utilizes the MediaPipe Face Mesh framework, which employs machine learning to estimate 468 3D facial landmarks from a single camera frame. This allows for high-precision tracking of the eyelids and facial boundaries without the need for depth sensors.

- Eye Aspect Ratio (EAR) Algorithm

To distinguish between open eyes and a blink, the system calculates the Eye Aspect Ratio (EAR). The EAR is the ratio of the vertical distance between the eyelids to the horizontal distance between the eye corners.

- Distance Estimation

Since the webcam has a fixed focal length, the pixel distance between the leftmost and rightmost points of the face (landmarks 234 and 454) increases as the user moves closer. The system monitors this width, if it exceeds a predefined pixel count, a "Too Close" warning is triggered.

Implementation Details

1. Software Stack

- Python: The core programming language for its robust image processing libraries.
- OpenCV: Used for video frame acquisition and rendering the Graphical User Interface (GUI).
- MediaPipe: Provides the deep learning model for real-time facial landmark detection.
- NumPy: Handles the vector mathematics required for EAR and distance calculations.

2. Operational Workflow

- Frame Capture: The webcam captures video at 30 frames per second.
- Pre-processing: Frames are converted to RGB and processed by the Face Mesh model.
- Analysis: The script extracts eye landmarks to calculate EAR and face width for distance.
- Thresholding: Logic checks if the user has blinked or is sitting too close.
- Output: Visual feedback is rendered on the screen in real-time.

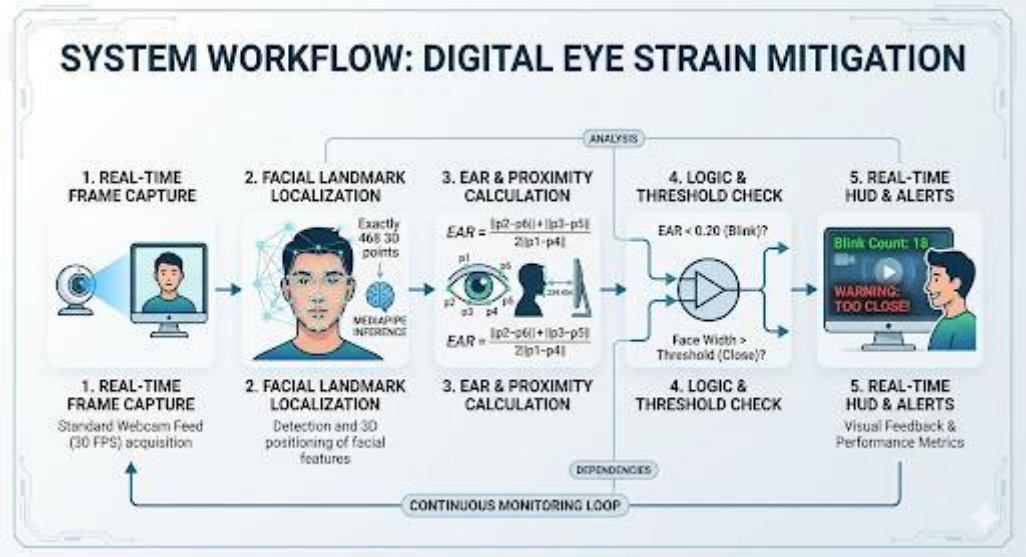


Fig 2. Workflow of the system

Results and Analysis

The system was tested in varying lighting conditions with the following observations:

• Accuracy: Blink detection achieved an accuracy of approximately 94% in well-lit environments.

• Performance: The system maintained a stable frame rate of 28-30 FPS on a standard Intel i5 processor, consuming less than 15% of CPU resources.

• User Interface: The Heads-Up Display provides immediate feedback, which has been shown in initial tests to effectively prompt users to blink or sit back.

Conclusion and Future Scope

The Digital Eye Strain Detection System successfully demonstrates how computer vision can be applied to preventive healthcare. It provides a functional, low-cost solution to a widespread modern problem using nothing more than a standard integrated webcam.

Future Enhancements:

• The 20-20-20 Rule: Integrating a timer that reminds users to look at something 20 feet away every 20 minutes.

• Data Logging: Developing a dashboard to track blink trends over weeks to provide long-term health insights.

• Auto-Brightness: Adjusting screen brightness based on detected eye strain levels.

References

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2.Souza, et al. (2023). "Real-Time Blink Detection as an Indicator of Computer Vision Syndrome in Real-Life Settings." MDPI.

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